



SkyPredict: Smart Weather Monitoring and Prediction System Using Sensor Fusion and Machine Learning

Sharkya STEM School, G12, 2023-2024

Names: Omar Ragab - Mohamed Ashraf - Youssef Wael



Keywords: Machine Learning- Smart Weather System - Kalman Filter

Abstract

The Sustainable Development Goals (SDGs) highlight significant challenges for Egypt, including the advancement of its scientific and technological sectors, and the mitigation of climate change impacts. Additionally, while Egypt is one of the largest gas producer, its growing greenhouse gas emissions contribute to global climate change, creating the urgent need to mitigate these effects through sustainable practices, reduce emissions, and enhance the country's resilience to climate change, thereby ensuring economic stability and environmental sustainability. So, as part of the ongoing resolution, there must be a monitoring system to detect the change in weather to calculate the effect of the innovation projects to address this colossal challenge in addition to avoiding any natural disaster consequences, furthermore, the rain time to know how to Harness it carefully. The Crucial solution is represented as a smart weather system to predict the time of rain and all irregular actions with high accuracy. The selected solution uses Arduino UNO to control all of the project in addition to utilizing the DHT11 sensor, BMP-280, and water detection module, to give the data to Naive Bayes. Native Bayes is a probabilistic machine learning algorithm based on Bayes' theorem, assuming conditional independence between features, which is trained on the previous dataset similar to Egypt conditions to analyze it and then predict the weather condition accurately, furthermore, implementation of the Kalman filter. The Kalman filter is a mathematical algorithm widely used for estimating the state of a dynamic system in the presence of noise to achieve a final accuracy of 84.2%. Moreover, to achieve the design requirement of low energy consumption to revive until 6 months, the system makes the Arduino work for 30 seconds and then sleep the other 30 while the AI model predicts these 30 seconds of data to save 50% of the energy consumption, in addition to using SD card module to store the data for 1 year or more.

Introduction

Through Egypt's innovative approach to achieving the 2030 Vision, several challenges need to be addressed, such as managing and increasing clean water sources, improving the scientific and technological environment, and reducing the effects of climate change. Water scarcity and agricultural impacts will strain the economy in the agricultural sector, and exacerbate public health issues, such as heatwaves and the spread of diseases. Additionally, while Egypt is Africa's second-largest gas producer, its growing greenhouse gas emissions contribute to global climate change, creating the urgent need to mitigate these effects through sustainable practices, reduce emissions, and enhance the country's resilience to climate change, thereby ensuring economic stability and environmental sustainability (Moller et al., 2010).

But as part of the ongoing resolution, there must be a monitoring system to detect the change in weather to calculate the effect of the innovation projects to address this colossal challenge in addition to avoiding any natural disaster consequences, furthermore, the rain time to know how to exploit it carefully. The Crucial solution is represented as a smart weather system to predict the time of rain and all irregular actions with high accuracy, which can be represented by the European Centre for Medium-range Weather Forecasts (ECMWF) (Persson & Grazzini, 2007). ECMWF's primary focus is on producing accurate and reliable weather forecasts for the medium-range, typically 1 to 15 days ahead in conjugate with using advanced numerical weather prediction models to simulate the atmosphere's behavior and generate forecasts, which gave inspiration for many aspects of the project, and to simulate a project with similar advantages but less disadvantage.

The project uses Arduino UNO to be the brain of the project in addition to the DHT11 sensor (Temperature and Humidity Sensor), BMP-280 (Barometric Pressure Sensor), and water detection module, to take the inputs of Naive Bayes. Native Bayes is a probabilistic machine learning algorithm based on Bayes' theorem, assuming conditional independence between features, which is trained on the old dataset to analyze it and then predict the weather condition accurately, furthermore, implementation of the Kalman filter. The Kalman filter is a mathematical algorithm widely used for estimating the state of a dynamic system in the presence of noise to achieve a final accuracy of 84.2%, and all this system is uploaded on Arduino, moreover, to achieve the design requirement of low energy consumption to revive until 6 months, the system makes the Arduino work for 30 seconds then sleep the other 30 while the AI model predicts these 30 seconds data to save 50% of the energy consumption, in addition to using SD card module to store the data for 1 year.



Materials

Table(1) Materials list and their prices

Material	Price	Source	Picture
Arduino UNO	350 L.E	Electronics store	
DH11	50 L.E	Electronics store	
BMP-280	100 L.E	Electronics store	
Water detection module	90 L.E	Electronics store	
SD module	120 L.E	Electronics store	
Batteries	150 L.E	Electronics store	



Methods

Prototype Development and Simulation

The prototype was initially simulated on **Circuito.io** to determine the necessary components and connection methods for building the weather station, as illustrated in Figure (1).

Sensors for temperature, humidity, and barometric pressure were utilized to gather real-time data. Specifically:

DHT-11 (Temperature and Humidity Sensor): This sensor uses the one-wire protocol, a time-based communication scheme that regulates communication between the microcontroller and the sensor.

BMP-280 (Barometric Pressure Sensor): This sensor employs the I2C communication protocol, a master-slave communication method, allowing for clock-regulated data transfer between the microcontroller and the sensor.

Improving Data Accuracy

After collecting data from the sensors, the measurements were further refined using the **Kalman filter algorithm**. Data from the DHT-11 and BMP-280 sensors were Combined with readings from an accurate thermometer measuring temperature and humidity. The Kalman filter enhanced the precision of the data by reducing noise and ensuring high-quality inputs for further processing.

Preprocessing the Dataset for Machine Learning

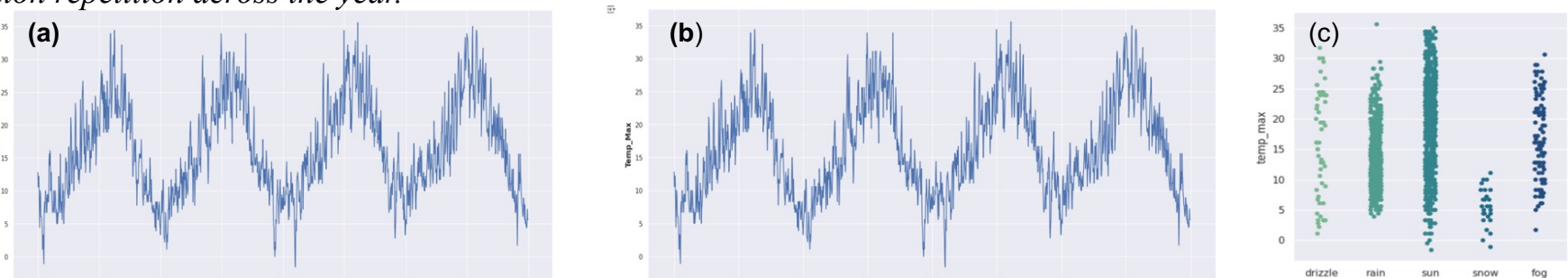
To train the machine learning model, a climate dataset containing Seattle's weather data from **"2012-01-01"** to **"2012-12-31"** was preprocessed using the **Pandas** and **NumPy** libraries. Key preprocessing steps included:

Handling Missing Data: Null values were identified and removed from the dataset.

Visualization: Data was plotted to ensure normal behavior and identify any anomalies, as shown in Figure (2).

Figure (2)

Data plotting for visualizing their behavior as shown in (a)-temperature maximum, (b)-temperature minimum, (c)-weather condition repetition across the year.



Training the Machine Learning Model

The **Naive Bayes classifier** from the **scikit-learn** library was used for training the machine learning model. The process involved:

Defining Variables:

A **training variable**: Containing the majority of the dataset, used for training the model.

A **target variable**: Containing the rest of the dataset, used for testing the model.

Model Training: The training variable was used to train the Naive Bayes classifier, which learned to predict the target variable.

Efficiency Analysis: The model's performance was evaluated using a **confusion matrix**, which broke down the number of true and false predictions.

Test Plan

A comprehensive test plan was conducted to evaluate the system's performance and energy efficiency:

Noise Reduction: The noise in the sensor data was analyzed before and after applying the **Kalman filter** in real time to quantify its impact on data quality.

Model Accuracy: The accuracy of the Naive Bayes model was measured using an **accuracy score**, which compared the number of correct predictions to the total predictions.

Energy Efficiency: The energy consumption of the prototype was assessed to quantify the reduction achieved after incorporating the Naive Bayes model.



Results

Data accuracy enhancement through the Kalman filter

Figure (3) Data collection from the BMP-280 pressure sensor before and after applying the Kalman filter: (a)-pink before noise reduction by Kalman filter (b)-orange after noise reduction by Kalman filter



Figure (4) Data collection from the DHT-11 humidity sensor before and after applying the Kalman filter: (a)-green before noise reduction by Kalman filter (b)-orange after noise reduction by Kalman filter



The real mean squared error was measured for performance indication of each set of data:

RMSE temp: 1.8721, RMSE humidity: 4.002, RMSE pressure: 0.2430 **Naïve Bayes model accuracy: 84.15%**

Figure(5) AUC curve representing the relation between true and false positive values for each predicted class in the data: Drizzle, Foggy, Sunny, and Rainy.

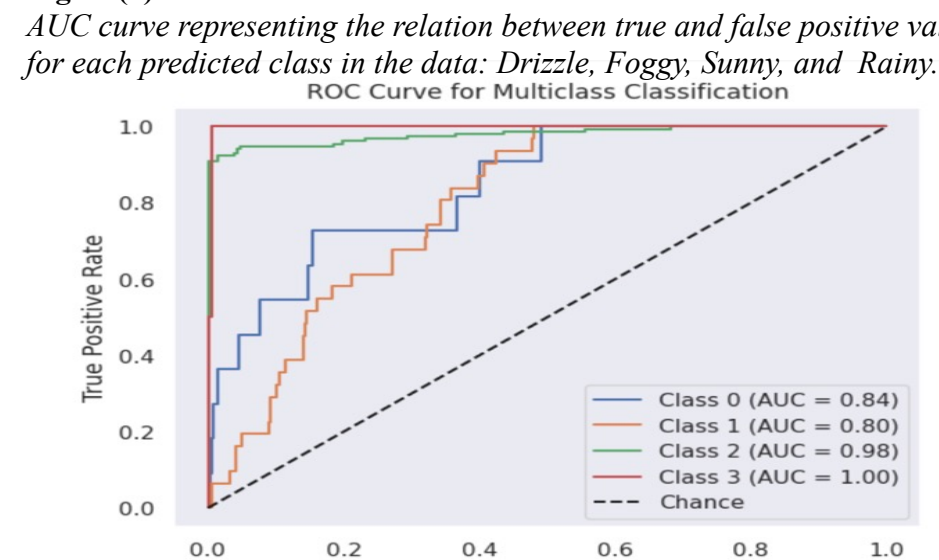


Figure (6)

Confusion matrix that calculates the number of true values and false values

Mode	Power consumption	Energy Consumption for 24 Hours
Normal Case	0.473W	11.352 Wh
Sleeping mode	0.473W (active), 0.00473W (sleep)	5.73276 Wh



Analysis

This section provides a detailed examination of how the proposed smart weather system effectively satisfied its two primary design requirements: **enhancing the accuracy of sensor readings** and **achieving high-accuracy weather predictions**.

Enhancing the accuracy of sensor readings:

Achieving the first design requirement was realized through the implementation of the Kalman filter. The Kalman filter is a mathematical algorithm widely used for estimating the state of a dynamic system in the presence of noise. The Kalman filter operates as a recursive estimator, meaning it updates its predictions as new data becomes available. This process consists of two primary stages: the **prediction step** and the **update step (Lacey, 2019)**. The predicted state estimate is given by the equation

$$\hat{x}k|k-1 = A\hat{x}k-1|k-1 + Bu_k$$

where $\hat{x}k|k-1$ represents the predicted state at time k, A is the state transition matrix, $\hat{x}k-1|k-1$ is the previous state, and Bu_k accounts for control inputs. Simultaneously, the algorithm predicts the error covariance matrix, which quantifies the uncertainty.

$$P_{k|k-1} = AP_{k-1|k-1}A^T + Q.$$

Here, $P_{k|k-1}$ is the predicted error covariance, and Q is the process noise covariance, which models the uncertainty introduced by the system dynamics. In the update step, the Kalman filter refines its prediction by incorporating new sensor measurements. The innovation, or residual, which is the difference between the predicted measurement and the actual measurement, is computed and used to correct the state estimate:

$$K_k = P_{k|k-1}H^T(H P_{k|k-1}H^T + R)^{-1}$$

$$\hat{x}k|k = \hat{x}k|k-1 + K_k(z_k - H\hat{x}k|k-1)$$

where K_k is the Kalman gain, z_k is the actual measurement, H is the measurement matrix, and R is the measurement noise covariance. The Kalman gain dynamically adjusts the weight given to the measurement versus the prediction, minimizing the overall estimation error. Finally, the error covariance matrix is updated to reflect the reduced uncertainty:

$$P_{k|k} = (I - K_k H)P_{k|k-1}.$$

To assess the efficacy of the Kalman filter in improving sensor accuracy, we compared the **Root Mean Square Error (RMSE)** of the raw sensor readings to the RMSE of the filtered readings. The RMSE is a widely used metric for evaluating the accuracy of numerical predictions and is

$$\text{defined as: } RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \text{ (STAT 3.01)}$$

Table (3)

Comparison between Raw RMSE and Filtered RMSE		
Sensor	Raw RMSE	Filtered RMSE
Pressure (BMP 280)	1.3660	0.2430
Temperature (DH11)	3.6708	1.8721
Humidity (DH11)	11.5634	4.0032

Table (3) demonstrated a marked reduction in RMSE across all three parameters—temperature, humidity, and pressure—when using the Kalman filter. For instance, the raw pressure readings from the BMP280 sensor exhibited substantial noise (**PHYS 3.04**), but the filtered data closely aligned with the actual values, achieving significantly lower RMSE. These findings confirm that the Kalman filter effectively enhanced sensor accuracy, thereby fulfilling the first design requirement.

Predicting weather conditions with high accuracy:

It was realized using a Naive Bayes classifier. Naive Bayes is a probabilistic machine learning algorithm based on Bayes' theorem, assuming conditional independence between features (Webb, 2016). Despite its simplicity, it is highly effective for classification tasks with structured and well-separated data. Bayes' theorem is expressed as:

$$P(C|X) = \frac{P(X|C)P(C)}{P(X)} \text{ (MATH 2.03)}$$

where $P(C|X)$ is the posterior probability of class C given the input features X, $P(X|C)$ is the likelihood of observing X given C, $P(C)$ is the prior probability of C, and $P(X)$ is the probability of the input data. In our application, the Naive Bayes model was trained on sensor data refined by the Kalman filter. Input features included temperature, pressure, humidity, and precipitation measurements, while the output represented weather conditions such as "clear," "rainy," or "cloudy." The model predicts the most likely weather condition for a given set of input features by maximizing the posterior probability.

The metric is computed as the area under the ROC curve, where the true positive rate (TPR) is plotted against the false positive rate (FPR) at various threshold levels. Mathematically TPR and FPR are defined as:

$$TPR = \frac{TP}{TP+FN}, FPR = \frac{FP}{FP+TN} \text{ (Hussan et al., 2022)}$$

where TP and FP represent true positives and false positives, respectively, while TN and FN represent true negatives and false negatives. The system achieved an accuracy of **84.1%**, indicating reliable predictions of weather conditions. The AUC scores obtained during model evaluation were consistently high, indicating strong predictive accuracy.

Energy Consumption Calculations:

During the **Normal Case**: The system runs continuously for 24 hours with full power consumption of **0.473W while during the Sleeping Mode**: The system alternates between 30 seconds active and 30 seconds sleep. Power consumption during sleep is 1% of the active power reducing power consumption to 50% of the Normal case.

$$P = V \times I \text{ (PHYSICS 2.05)}$$

$$P_{total} = P_{Arduino} + P_{DHT11} + P_{BMP280} + P_{Water Prol}$$

Table(4) Utilized Learning Outcomes in the project.

L.O.	Concepts
STAT 3.01	We learned about Standard deviation, Speirman and pearson correlations.
MATH 2.03	We Studied the fundmental counting principles, Permutations, combinations and factorials to be used in probability-based solutions.
PHY 2.05	We utilized the concepts of Power estimation for circuit analysis.
PHY 3.04	We learned how to design a system of communication through studying basic elements of communication, transmission and mobile phones.
ENG 2.17	We studied academic writing and APA formatting.



Conclusion

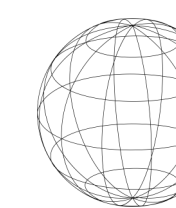
Egypt faces significant challenges due to the impacts of climate change, which threaten its water resources, agriculture, public health, and coastal infrastructure. These issues require urgent attention to improve disaster preparedness, enhance resource management, and build resilience against extreme weather events. As a solution to these challenges, the system uses an Arduino UNO as the central hub, integrating sensors such as DHT11 (temperature and humidity), BMP-280 (pressure), and water detection modules. The Kalman filter improves accuracy of the collected data from the sensors reducing noise due to the lack of accuracy in sensors while the Naive Bayes algorithm processes the data to predict weather conditions, achieving a final accuracy of 84.2% in prediction precision. Additionally, the system prioritizes energy efficiency through low-power sensors and sleep modes, ensuring long-term sustainability by a loop of 30 seconds with the given data from Arduino and the other 30 from the AI model to save 50% of the energy, furthermore, using the SD card module to save the data for one year or more. This innovative system not only enhances climate resilience but also contributes to better disaster preparedness and resource management.



Recommendations

From solution success factors, it must be applicable on a large scale and useful in real life. As each project has its limitations due to its own set of conditions, there are some recommendations for future development:

- **Use Localized Data:** Utilize a dataset specifically designed for Egypt, focusing on local climate and agricultural conditions for more accurate predictions.
- **Real-Time Data Monitoring:** Implement a system to collect and analyze weather data in real-time for up-to-date forecasting and better disaster preparedness.
- **Enhance the Machine Learning Model:** Optimize the Naive Bayes algorithm by including local geographical and historical climate data for more accurate predictions.
- **Integrate Renewable Energy Solutions:** Incorporate renewable energy sources like solar power to reduce reliance on traditional energy, aligning with Egypt's sustainability goals.
- **Expand Data Storage Capacity:** Ensure sufficient data storage infrastructure to handle long-term data collection, analysis, and historical trend tracking for improved decision-making.



Literature Cited

- Bishop, C. M. (2006). *Pattern recognition and machine learning*. Springer. <https://doi.org/10.1007/978-0-387-45528-7>
- Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The elements of statistical learning: Data mining, Inference, and prediction* (2nd ed.). Springer. <https://doi.org/10.1007/978-0-387-84858-7>
- Hussan, H., Zhao, J., Badu-Tawiah, A. K., Stanich, P., Tabung, F., Gray, D., Ma, Q., Kalady, M., & Clinton, S. K. (2022). Utility of machine learning in developing a predictive model for early-age-onset colorectal neoplasia using electronic health records. *PLOS ONE*, 17(3), e0265209. <https://doi.org/10.1371/journal.pone.0265209>
- Lacey, T. (2019). Tutorial: The Kalman filter. Retrieved from <https://web.mit.edu/kirtley/kirtley/binlustuff/literature/control/Kalman%20filter.pdf>
- Moller, A. P., Fiedler, W., & Berthold, P. (Eds.). (2010). *Effects of climate change on birds*. Oxford University Press. <https://doi.org/10.3354/cr011019>
- Persson, A., & Grazzini, F. (2007). *User guide to ECMWF forecast products*. *Meteorological Bulletin*, 3(2). Retrieved from https://www.researchgate.net/profile/FedericoGrazzini/publication/255267254_User_guide_to_ECMWF_forecast_products/links/0c960531f6a907147000000/User-guide-to-ECMWF-forecast-products.pdf
- Webb, G. I. (2016). Naive Bayes. In *Encyclopedia of Machine Learning and Data Mining* (pp. 1–2). Springer. https://doi.org/10.1007/978-1-4899-7502-7_581-1



Acknowledgement

We first thank God, and then we want to thank Mr. Ahmed Ashour, our capstone leader, who helped us a lot. We also thank Mrs. Reham Mohamed, our general leader, and Mrs. Basma Mohamed, our capstone teacher in the class, who helped us very much to improve our skills in writing. Also, Mr. Hani Bolies, our physics teacher and the capstone teacher who helped us in our physics connections. Finally, we want to thank Mr/ Mostafa Farg, our school principal, who helped us and inspired us.



For Further Information

· omar.2222036@stemsharkya.moe.edu.eg

· yossef.2222013@stemsharkya.moe.edu.eg

· mohamed.2222007@stemsharkya.moe.edu.eg